WEEK 3 · LECTURE L2 OF 3 · UNIT I

Business Intelligence Fundamentals

What a warehouse, a metric and a dashboard actually are, and why GreenLeaf has no single source of truth.

Part 1 Sources, warehouse, metric layer, dashboard

Part 2 The single source of truth

Part 3 Exercise: measuring time to hire honestly

Read Caldwell (2022), BI foundations

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PART 1 · BI FUNDAMENTALS

GreenLeaf's data, mapped onto a BI stack

BI layer	What it does	What GreenLeaf has
Sources	Systems where work happens	LegacyHR, PeopleCloud, HireRight ATS, GreenLearn LMS, ESS App, PayEasy
Pipeline	Extracts, cleans and loads data	None: data is re-keyed by hand into spreadsheets
Warehouse	Keeps history in one place	None
Metric layer	Agreed definitions with owners	None: each team defines its own
Dashboard	Shows decisions, not data	InsightHR Analytics, used by 38 of 90 licensed users (0.42)

THE GAP GreenLeaf bought a dashboard before building the layers underneath it. Source: TechStack_Inventory.

BI FUNDAMENTALS

A warehouse is a copy, kept on purpose

THE SOURCES

Operational systems

Payroll, the HRIS, the ATS, the learning platform. Each built to *run* a process, and optimized for today's transaction rather than for last year's history.

THE PIPELINE

Extract, conform, load

Pull the data out, force it onto shared definitions and shared keys, write it down. **The conforming step is where the politics lives.**

THE WAREHOUSE

History that does not move

A separate store, organized for asking questions rather than for running payroll, that remembers what was true in March even after March has been overwritten.

The thing nobody tells you

Operational systems mostly hold the **present**. When a person changes grade, the field is overwritten and the old value is gone. A warehouse exists largely to keep the past — and without it, no trend, no cohort, no predictive model is possible.

BI FUNDAMENTALS

A metric is an agreement, not a formula

“Attrition rate” is not a definition. It becomes one only when six further questions have been answered and written down.

Numerator
All leavers, or voluntary only? Retirements? End of fixed term?

Denominator
Opening headcount, closing, or average? Contractors in or out?

Period
Financial year or rolling twelve months? Annualized from a quarter?

Grain
Whole firm, department, site, manager? At what size do you suppress?

Owner
Who may change the definition, and who must be told when it changes?

Restatement
When it changes, is history recalculated or left as published?

Two teams using different answers to any one of these produce different numbers, both correct. **That is not a data-quality problem. It is an absent agreement** — and it cannot be fixed by buying software.

BI FUNDAMENTALS

A dashboard is a decision surface

Not a report, and not a place to display everything you happen to have. The test of a dashboard is whether someone changes what they do after looking at it.

Every tile needs an owner and an action

If nobody is accountable for the number and nothing follows from it moving, the tile is decoration that costs attention.

Comparison beats level

A bare number cannot be interpreted. Against last period, against a target, against a peer unit — the comparison is what carries the meaning.

A number that is watched gets managed

Publish time-to-fill and time-to-fill improves — sometimes by better hiring, sometimes by opening requisitions later. Measurement changes behavior.

Goodhart's observation

When a measure becomes a target, it stops being a good measure. **Choosing what to put on a dashboard is choosing what the organization will optimize** — including in ways you did not intend.

L2 · BI FUNDAMENTALS

A single source of truth, and why GreenLeaf has none

What it means

Every report draws on one agreed record for each employee. Ask two managers for headcount and you get the same number.

Two core systems

LegacyHR (on premises, adoption 0.43) and PeopleCloud (pilot, adoption 0.38) both hold employee records.

The spreadsheet paradox

Spreadsheet Trackers: adoption 1.00, satisfaction 1.5 out of 5 (the lowest), zero integrations.

READ THE PARADOX

Universal use with the lowest satisfaction is not success. It is the fallback people use because no system is trusted. Source: TechStack_Inventory.

PART 2 · DATA QUALITY

Real HR data is never clean, and GreenLeaf's is no exception

Impossible records

139 learning records marked Not-started still carry an assessment score. Rule: treat Not-started as 0.

Records that disagree

A few employees' Age and Tenure_Months do not reconcile. Rule: treat Age as authoritative.

Labels that do not match

Some Primary_System entries are composites like "PayEasy + Sheets" with no match in the stack list.

THE HABIT

Write down every cleaning rule you apply. A figure nobody can reproduce is an opinion. Source: the Data Dictionary, integrity notes.

L2 · EXERCISE

Sketch the data you need to measure time to hire honestly

Start the clock

When the vacancy opens, when it is approved, or when it is advertised? Each choice gives a different number.

Stop the clock

Offer made, offer accepted, or first day at work? GreenLeaf's ATS stops at the first joiner's start date.

Choose the summary

Mean or median? Across 140 requisitions the mean is 43.8 days and the median 43. One outlier can move a mean.

Split it

By role, by Location_Tier, by Source_Channel. A single average can hide a slow Tier 3 store channel.

Agree it

Write the definition down and give it an owner. A metric is an agreement, not a formula.

TRY IT

Recruitment_ATS, column Time_to_Fill_Days. Then ask why the Board briefing quotes 47 days for frontline roles.

Going deeper: the theory underneath business intelligence

The next slides set out Ackoff's hierarchy, Inmon and Kimball, Goodhart and Campbell, and the dimensions of data quality: what each claims, how it works, where it breaks, and what it means at GreenLeaf.

THEORY IN DEPTH · ACKOFF (1989)

From data to wisdom

THE CORE IDEA

Data are symbols; information answers who, what, where; knowledge answers how; wisdom judges what is worth doing.

HOW IT WORKS

Each level adds context and purpose. A dashboard sits at information; a decision needs knowledge.

LIMITS AND CRITIQUE

The neat pyramid is criticized: in practice the levels blur, and judgment shapes even raw data.

AT GREENLEAF

GreenLeaf has data in spreadsheets but little information, because nobody has agreed what the numbers mean.

THEORY IN DEPTH · INMON (1992); KIMBALL (1996)

Two ways to build a warehouse

THE CORE IDEA

Inmon: build one integrated enterprise warehouse first, then marts. Kimball: build business-focused marts with shared dimensions.

HOW IT WORKS

Kimball's star schema joins a fact table (events) to dimension tables (who, when, where).

LIMITS AND CRITIQUE

Inmon is slower but consistent; Kimball is faster but needs discipline on shared definitions.

AT GREENLEAF

PeopleAnalytics_Fact is a fact table; Employee_Master acts as a dimension. That is a Kimball design.

THEORY IN DEPTH · GOODHART (1975); CAMPBELL (1976)

A measure that becomes a target stops measuring

THE CORE IDEA

When people are rewarded on a number, they change their behavior to move the number.

HOW IT WORKS

Campbell: the more a quantitative indicator drives social decisions, the more it is corrupted.

LIMITS AND CRITIQUE

Not every target fails; pairing a metric with a counter-metric limits gaming.

AT GREENLEAF Reward recruiters on time to hire, and quality of hire may fall. Track both.

THEORY IN DEPTH · WANG AND STRONG (1996)

Data quality has four families

THE CORE IDEA

Quality is fitness for use, judged by the people who use the data.

HOW IT WORKS

Intrinsic (accuracy), contextual (relevance, timeliness), representational (clarity), accessibility (can it be reached).

LIMITS AND CRITIQUE

The dimensions overlap, and fitness depends on the task, so no single score fits all uses.

AT GREENLEAF

139 learning records marked Not-started with a score fail intrinsic quality. Composite system labels fail representational quality.

BRIDGE TO L3

The same machinery that answers questions can watch people

1

A warehouse keeps history on purpose. That is its power and its risk.

2

A metric is an agreement with an owner, not a formula in a cell.

3

A dashboard should change a decision. If it changes nothing, it is decoration.

NEXT Who consented to being measured, and what are they owed? The ethics of adoption.